Assignment: Definite Integrals

ISI UGB Mastery

Part I: Problems

Problem 1

Let $f\colon [0,1] \to \mathbb{R}$ be a continuously differentiable function satisfying $f(0) = 0$ and $f(1) = 1$. Prove that

$$\int_0^1 |f'(x) - f(x)|\, dx \;\geq\; \frac{1}{e}.$$

Problem 2

Evaluate the following limit rigorously:

$$\lim_{n\to\infty} \sum_{k=1}^{n} \left(\frac{k}{n}\right)^{n}.$$

Problem 3

Evaluate the following definite integral involving the floor function:

$$\int_0^1 (-1)^{\lfloor 1/x \rfloor}\, dx.$$

Problem 4

Let $f\colon [0,1] \to \mathbb{R}$ be a continuous function such that for every integer k with $0 \leq k \leq n$,

$$\int_0^1 x^k f(x)\, dx = 0.$$

Prove that f has at least $n+1$ distinct roots in the open interval $(0,1)$.

Problem 5

Let $f\colon [0,1] \to \mathbb{R}$ be a continuously differentiable function with $f(0) = f(1) = 0$. Prove that

$$\int_0^1 (f'(x))^2\, dx \;\geq\; 12\left(\int_0^1 f(x)\, dx\right)^2.$$

Problem 6

Let $f\colon [0,1] \to \mathbb{R}$ be a continuously differentiable function. Evaluate, in terms of $f(1)$, the limit

$$\lim_{n\to\infty} n^2 \int_0^1 x^n (1-x)\, f(x)\, dx.$$

Problem 7

Let $f\colon [0,1] \to \mathbb{R}$ be a twice continuously differentiable function with $f(0) = f(1) = 0$, satisfying

$$\int_0^1 (f''(x))^2\, dx = 1.$$

(a) Prove that

$$\left| \int_0^1 f(x)\, dx \right| \le \frac{1}{2\sqrt{30}}.$$

(b) Exhibit an explicit function f for which equality holds.

Part II: Solutions

Solution to Problem 1

The $|f'(x) - f(x)|$ inside the integral is the hint. Whenever we see $f' - f$, we should think *integrating factor* — because the combination $f' - f$ is exactly what pops out when we differentiate $e^{-x}f(x)$. Let's check:

$$\frac{d}{dx}\big(e^{-x}f(x)\big) = e^{-x}f'(x) - e^{-x}f(x) = e^{-x}\big(f'(x) - f(x)\big).$$

Now $e^{-x} \le 1$ on $[0, 1]$, which means

$$|f'(x) - f(x)| \;\ge\; e^{-x}\,|f'(x) - f(x)| = \left|\frac{d}{dx}\big(e^{-x}f(x)\big)\right|.$$

Integrate both sides from 0 to 1:

$$\int_0^1 |f'(x) - f(x)|\,dx \;\ge\; \int_0^1 \left|\frac{d}{dx}\big(e^{-x}f(x)\big)\right| dx.$$

The right side has absolute value *inside* the integral; we want to pull it outside to turn the integral into something concrete. That's exactly what the triangle inequality for integrals lets us do:

$$\int_0^1 \left|\frac{d}{dx}\big(e^{-x}f(x)\big)\right| dx \;\ge\; \left|\int_0^1 \frac{d}{dx}\big(e^{-x}f(x)\big)\,dx\right|.$$

Now the integral inside collapses by the Fundamental Theorem of Calculus:

$$\int_0^1 \frac{d}{dx}\big(e^{-x}f(x)\big)\,dx = e^{-1}f(1) - f(0) = \frac{1}{e} - 0 = \frac{1}{e}.$$

Putting everything together gives $\int_0^1 |f' - f|\,dx \ge 1/e$. ■

Solution to Problem 2

The sum $S_n = \sum_{k=1}^{n}(k/n)^n$ has terms that are tiny for small k (since $(k/n)^n$ is a small number raised to a large power) and close to 1 when k is close to n. So the real action happens at the top of the sum. Let's flip the index to bring that action to the front: set $j = n - k$, which runs from 0 to $n - 1$ as k runs from n down to 1. Then

$$S_n = \sum_{j=0}^{n-1}\left(1 - \frac{j}{n}\right)^n.$$

Now we can read off what each term looks like for large n. For a fixed j, the standard limit

$$\left(1 - \frac{j}{n}\right)^n \;\longrightarrow\; e^{-j}$$

suggests that $S_n \to \sum_{j=0}^{\infty} e^{-j}$. But we can't just wave our hand and swap the limit with the sum — we need a reason. The way to do it is to notice that

$$\left(1 - \frac{j}{n}\right)^n \;\le\; e^{-j} \quad \text{for } 0 \le j \le n,$$

which follows from the inequality $\ln(1-t) \le -t$ for $t \in [0,1)$. So each term in our sum is bounded above by e^{-j}, and $\sum_{j=0}^{\infty} e^{-j}$ is a convergent geometric series. That's the dominating series we need.

By the Dominated Convergence Theorem (in its counting-measure form, which is the same as the Weierstrass M-test), we can interchange the limit and the sum:

$$\lim_{n\to\infty} S_n = \sum_{j=0}^{\infty} e^{-j} = \frac{1}{1-1/e} = \frac{e}{e-1}.$$

Solution to Problem 3

The $\lfloor 1/x \rfloor$ in the exponent is scary-looking but has a simple structure. For x in the interval $\left(\frac{1}{k+1}, \frac{1}{k}\right]$, the reciprocal $1/x$ lives in $[k, k+1)$, so $\lfloor 1/x \rfloor = k$. The interval $(0,1]$ breaks into infinitely many of these pieces, and on each piece the integrand is just the constant $(-1)^k$. So we can split the integral:

$$I = \int_0^1 (-1)^{\lfloor 1/x \rfloor}\,dx = \sum_{k=1}^{\infty} (-1)^k \int_{1/(k+1)}^{1/k} dx = \sum_{k=1}^{\infty} (-1)^k \left(\frac{1}{k} - \frac{1}{k+1}\right).$$

Now split this into two sums:

$$I = \sum_{k=1}^{\infty} \frac{(-1)^k}{k} - \sum_{k=1}^{\infty} \frac{(-1)^k}{k+1}.$$

The first sum is the alternating harmonic series. Its value is $-\ln 2$ (it's minus the standard $\ln 2 = 1 - 1/2 + 1/3 - \cdots$).

For the second sum, shift the index by letting $j = k+1$, so when $k = 1$ we have $j = 2$:

$$\sum_{k=1}^{\infty} \frac{(-1)^k}{k+1} = \sum_{j=2}^{\infty} \frac{(-1)^{j-1}}{j} = -\sum_{j=2}^{\infty} \frac{(-1)^j}{j}.$$

The sum $\sum_{j=1}^{\infty} (-1)^j/j = -\ln 2$, so dropping the $j = 1$ term (which equals -1) gives $\sum_{j=2}^{\infty} (-1)^j/j = -\ln 2 - (-1) = 1 - \ln 2$. So the second sum equals $-(1 - \ln 2) = \ln 2 - 1$.

Putting it together:

$$I = -\ln 2 - (\ln 2 - 1) = 1 - 2\ln 2.$$

Solution to Problem 4

We'll argue by contradiction. Suppose, for the sake of argument, that f changes sign at only m points of $(0,1)$, with $m \le n$. Call these points $c_1 < c_2 < \cdots < c_m$.

Now here's the idea: build a polynomial that has exactly those same sign changes. Let

$$P(x) = (x - c_1)(x - c_2)\cdots(x - c_m).$$

Because P switches sign at exactly the same places as f inside $(0,1)$, the product $f(x)P(x)$ never flips sign — it stays on one side of zero throughout $(0,1)$. Since f is continuous and not identically zero, this product isn't identically zero either. So

$$\int_0^1 f(x)P(x)\,dx \ne 0.$$

But wait — $P(x)$ has degree $m \le n$. That means we can write

$$P(x) = a_0 + a_1 x + a_2 x^2 + \cdots + a_m x^m,$$

with each power of x being at most n. By linearity of the integral, and using the hypothesis that $\int_0^1 x^k f(x)\,dx = 0$ for every k from 0 to n:

$$\int_0^1 f(x)P(x)\,dx = \sum_{k=0}^{m} a_k \underbrace{\int_0^1 x^k f(x)\,dx}_{=\,0} = 0.$$

So the integral is both nonzero and zero at the same time — contradiction. Our assumption that f has only $m \le n$ sign changes must be wrong, and so f changes sign (and therefore has roots) at least $n+1$ times inside $(0,1)$. ■

Solution to Problem 5

Let $I = \int_0^1 f(x)\,dx$. We want to connect this to f', and the natural way is integration by parts. Write $I = \int_0^1 1 \cdot f(x)\,dx$ and choose $dv = dx$, $u = f(x)$.

Now, the naive choice is $v = x$. But look at our boundary conditions: $f(0) = f(1) = 0$. Both endpoints vanish, and the symmetry of the problem suggests choosing an antiderivative that treats the endpoints equally. So instead of $v = x$, we take

$$v = x - \tfrac{1}{2}$$

(still a valid antiderivative of 1; the constant doesn't matter). Integration by parts gives

$$I = \left[\left(x - \tfrac{1}{2}\right) f(x)\right]_0^1 - \int_0^1 \left(x - \tfrac{1}{2}\right) f'(x)\,dx.$$

Because f vanishes at both endpoints, the boundary term drops out:

$$I = -\int_0^1 \left(x - \tfrac{1}{2}\right) f'(x)\,dx.$$

Now we have I written as an integral of a product involving f'. That's the setup for Cauchy–Schwarz:

$$I^2 \;\le\; \left(\int_0^1 \left(x - \tfrac{1}{2}\right)^2 dx\right)\left(\int_0^1 (f'(x))^2\,dx\right).$$

The first integral is a direct computation:

$$\int_0^1 \left(x - \tfrac{1}{2}\right)^2 dx = \left[\frac{(x - 1/2)^3}{3}\right]_0^1 = \frac{1/8 - (-1/8)}{3} = \frac{1}{12}.$$

Plugging in:

$$I^2 \;\le\; \frac{1}{12}\int_0^1 (f'(x))^2\,dx \;\implies\; \int_0^1 (f'(x))^2\,dx \;\ge\; 12\,I^2 = 12\left(\int_0^1 f(x)\,dx\right)^2.$$

■

Solution to Problem 6

The factor x^n inside the integral becomes sharply peaked near $x = 1$ as n grows — outside a tiny neighbourhood of 1, x^n is practically zero. So all the mass of the integral concentrates near $x = 1$, and the rest of the integrand ($f(x)$ and $(1-x)$) is smooth there. The natural move is to shift that neighbourhood to the origin, where it's easier to analyse.

Substitute $x = 1 - u/n$, so $dx = -du/n$. When $x = 0$, $u = n$; when $x = 1$, $u = 0$. So

$$\int_0^1 x^n(1-x)f(x)\,dx = \int_n^0 \Big(1-\frac{u}{n}\Big)^n \frac{u}{n}\, f\Big(1-\frac{u}{n}\Big)\Big(-\frac{du}{n}\Big) = \frac{1}{n^2}\int_0^n \Big(1-\frac{u}{n}\Big)^n u\, f\Big(1-\frac{u}{n}\Big)\,du.$$

(The minus sign gets absorbed when we flip the integration limits.) Multiplying by the n^2 from the original limit cancels the $1/n^2$, leaving

$$n^2\int_0^1 x^n(1-x)f(x)\,dx = \int_0^n \Big(1-\frac{u}{n}\Big)^n u\, f\Big(1-\frac{u}{n}\Big)\,du.$$

Now look at what happens to each piece as $n \to \infty$ (for a fixed u):

- $(1-u/n)^n \to e^{-u}$ (the standard exponential limit);
- $f(1-u/n) \to f(1)$ (since f is continuous);
- the upper limit n goes to infinity.

So the integral looks like it should tend to $\int_0^\infty e^{-u}\,u\,f(1)\,du$. To justify taking the limit inside the integral, we need a dominating function. Since f is continuous on $[0,1]$, it's bounded by some M; and $(1-u/n)^n \le e^{-u}$ for $0 \le u \le n$. So the integrand is bounded above by $M\,u\,e^{-u}$, which is integrable on $[0,\infty)$.

By the Dominated Convergence Theorem,

$$\lim_{n\to\infty} n^2\int_0^1 x^n(1-x)f(x)\,dx = f(1)\int_0^\infty u\,e^{-u}\,du.$$

The last integral is a standard gamma integral: $\int_0^\infty u\,e^{-u}\,du = \Gamma(2) = 1$. So the answer is $f(1)$.

Solution to Problem 7

(a) Let $I = \int_0^1 f(x)\,dx$. We're given information about f'' and we want to bound something about f. The way to connect them is integration by parts — twice. But we need to be careful: the boundary terms from integration by parts are going to bite us unless we set things up carefully.

Here's the idea. We'll write $I = \int_0^1 1\cdot f(x)\,dx$, and then look for a function $P(x)$ satisfying $P''(x) = 1$. If we integrate by parts twice using P as our antiderivative chain, each step will produce boundary terms of the form $[P'\cdot f]_0^1$ and $[P\cdot f']_0^1$. To kill *both* of these, we want $P(0) = P(1) = 0$.

Solving $P'' = 1$ with $P(0) = P(1) = 0$ gives

$$P(x) = \frac{x^2 - x}{2}.$$

(Integrate $P'' = 1$ twice to get $P(x) = \frac{1}{2}x^2 + C_1 x + C_2$, then the two boundary conditions pin down $C_1 = -1/2$, $C_2 = 0$.)

Now rewrite the integral using $P'' = 1$:

$$I = \int_0^1 P''(x)\, f(x)\, dx.$$

First integration by parts (boundary term vanishes because $f(0) = f(1) = 0$):

$$I = \big[P'(x)f(x)\big]_0^1 - \int_0^1 P'(x)f'(x)\, dx = -\int_0^1 P'(x)f'(x)\, dx.$$

Second integration by parts (boundary term vanishes because $P(0) = P(1) = 0$):

$$I = -\big[P(x)f'(x)\big]_0^1 + \int_0^1 P(x)f''(x)\, dx = \int_0^1 P(x)f''(x)\, dx.$$

Now we have I as an integral of a product involving f'', and Cauchy–Schwarz takes it from here:

$$I^2 \;\le\; \left(\int_0^1 P(x)^2\, dx\right)\left(\int_0^1 (f''(x))^2\, dx\right) = \int_0^1 P(x)^2\, dx,$$

using the hypothesis $\int_0^1 (f'')^2 = 1$. Now compute:

$$\int_0^1 \left(\frac{x^2 - x}{2}\right)^2 dx = \frac{1}{4}\int_0^1 (x^4 - 2x^3 + x^2)\, dx = \frac{1}{4}\left(\frac{1}{5} - \frac{1}{2} + \frac{1}{3}\right) = \frac{1}{4}\cdot\frac{1}{30} = \frac{1}{120}.$$

So $I^2 \le 1/120$, which gives $|I| \le 1/\sqrt{120} = 1/(2\sqrt{30})$.

(b) To hit equality in Cauchy–Schwarz, the two functions in the inequality must be proportional: $f''(x) = \lambda\, P(x)$ for some constant λ. We pin down λ using the constraint $\int_0^1 (f'')^2 = 1$:

$$\lambda^2 \int_0^1 P(x)^2\, dx = 1 \implies \lambda^2 \cdot \frac{1}{120} = 1 \implies \lambda = \pm 2\sqrt{30}.$$

Pick $\lambda = 2\sqrt{30}$ (the other sign gives $-f$, which also works). Then

$$f''(x) = \sqrt{30}\,(x^2 - x).$$

Integrate twice:

$$f'(x) = \sqrt{30}\left(\frac{x^3}{3} - \frac{x^2}{2} + C\right), \qquad f(x) = \sqrt{30}\left(\frac{x^4}{12} - \frac{x^3}{6} + Cx + D\right).$$

Use the boundary conditions to find the constants: $f(0) = 0$ gives $D = 0$; then $f(1) = 0$ gives $\frac{1}{12} - \frac{1}{6} + C = 0$, so $C = \frac{1}{12}$. Substituting back:

$$f(x) = \frac{\sqrt{30}}{12}\,(x^4 - 2x^3 + x).$$

■